WNBA Analysis Project: Model Benchmarks & 2027 Season Projections

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Introduction

This is the WNBA analog of `NBA_Analysis_Project.Rmd` (T2 / `wnba_rating_engine_agent`, Phase 2 of `task_assignments.md`). Per `WNBA_REPLICATION_PLAN.md` §(b)/(d) and the agent brief, this is **not** a find-and-replace port of the NBA Rmd - the same methodology (rolling-window feature comparison, offseason-transition comparison, model architecture benchmark, Monte Carlo season simulation) is re-run from scratch on the real WNBA data foundation `daily_ingest.R/player_metrics.R` (T1) already built, using `wehoop` team box scores (2019-2026, `box_scores_cache.rds`, 3,906 real team-game rows across the 15 real `WNBA_TEAMS` franchises - see `task_assignments.md`'s T0/T1 Handoff Notes for the full provenance/validation history of that cache).

Key differences from the NBA baseline that this analysis explicitly re-tests rather than assumes:

1. **Rolling window size**: NBA used 18 games out of an 82-game season (~22%). `config.R` currently sets `ROLLING_WINDOW_SIZE <- 9` as a *proportional placeholder* (18 * 44/82 ~= 9.66 -> 9) - this notebook actually re-runs the NBA Rmd's own "test window sizes" experiment on real WNBA data (5/7/9/11/13 games) rather than trusting the proportional-scaling assumption blindly.
2. **Season length**: `WNBA_SEASON_GAMES <- 44` (not 82) drives the Monte Carlo schedule simulation (round-robin games-per-opponent, zero-sum win totals, ridge-plot win-total axis).
3. **Offseason transition method**: `config.R`'s `OFFSEASON_METHOD <- "reset"` was carried over from NBA as a *starting default*, flagged as needing re-benchmarking because a ~44-game season accumulates less in-season signal before cross-season noise dominates than NBA's 82-game season - continuity/decay methods may outperform "reset" more often here. Re-tested below.
4. **Team universe**: 15 real WNBA franchises (`config.R`'s `WNBA_TEAMS`, including the 2026 expansion Toronto Tempo and Portland Fire franchises) instead of NBA's 30.

IMPORTANT STATUS NOTE (read before trusting any number below): every code chunk in this file is real, complete, and was actually run once by the authoring agent via `Rscript.exe` against the live caches to sanity-check it executes end-to-end on real data (see `task_assignments.md`'s T2 Handoff Notes for the exact numbers that run produced). However, **this .Rmd file itself has not been formally `knitr::knit()`/`rmarkdown::render()`-ed to produce a PDF** as part of this task (no PDF renderer/TeX toolchain dependency was verified in this environment) - so the numbers embedded in this document as prose commentary should be treated as "verified reproducible via direct script execution, but the Rmd artifact itself is unexecuted/unknit." Re-run `rmarkdown::render("WNBA_Analysis_Project.Rmd")` end-to-end before treating this as the new pristine baseline analog to the NBA Rmd (per `ADR.md` Principle I / this agent's "pristine baseline" discipline noted in `agent3.md`).

Load packages:

```
library(tidyverse)
library(tidymodels)
library(wehoop)
library(slider)
```

```r
library(xgboost)
library(ggridges)
source("config.R")
```

Data Ingestion & Caching

Load the box score data from the `wehoop` API for seasons 2019 through 2026 (via T1's `daily_ingest.R`-produced cache, which already applies the `CONN->CON` normalization and the All-Star/exhibition-row filter documented in `task_assignments.md`'s T0/T1 Handoff Notes):

```r
if (file.exists(CACHE_TEAM_BOX)) {
  box_scores <- readRDS(CACHE_TEAM_BOX)
} else {
  box_scores <- wehoop::load_wnba_team_box(seasons = ALL_SEASONS) |>
    filter(team_abbreviation %in% names(WNBA_TEAMS))
  saveRDS(box_scores, CACHE_TEAM_BOX)
}

cat("Total team box score records:", nrow(box_scores), "\n")
```

```
## Total team box score records: 3906
```

```r
cat("Distinct franchises:", n_distinct(box_scores$team_abbreviation), "\n")
```

```
## Distinct franchises: 15
```

```r
cat("Seasons covered:", paste(sort(unique(box_scores$season)), collapse = ", "), "\n")
```

```
## Seasons covered: 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026
```

Feature Engineering & Offseason Transition Experiments

We compute team rolling averages for key performance indicators (eFG%, Turnover Rate, Free Throw Rate, True Shooting %, Points, Rebounds, Assists, Steals, Blocks) - identical feature definitions to the NBA baseline (none of these ratios are game-length- or roster-size-dependent, so no formula changes were needed here per `WNBA_REPLICATION_PLAN.md`'s Tier 1/2 classification).

To resolve offseason transition artifacts, we test the same three methodologies as the NBA baseline: - **Method A (Fresh Reset):** Rolling windows reset cleanly at game 1 of each new season. - **Method B (50% Regression to Mean):** New-season baselines regress 50% toward the league average. - **Method C (Exponential Decay Carryover):** Rolling stats carry over across seasons via an 80/20 blend.

```r
build_feature_df <- function(data, window_size = ROLLING_WINDOW_SIZE, method = "reset") {
  if (method == "reset") {
    features <- data |>
      arrange(team_id, season, game_date) |>
      group_by(season, team_id) |>
      mutate(
        efg = (field_goals_made + 0.5 * three_point_field_goals_made) / pmax(1, field_goals_attempted),
        tov = total_turnovers / pmax(1, field_goals_attempted + 0.44 * free_throws_attempted + total_tu
        ftr = free_throws_attempted / pmax(1, field_goals_attempted),
        ts  = team_score / pmax(1, 2 * (field_goals_attempted + 0.44 * free_throws_attempted)),

        roll_pts_scored = slide_dbl(team_score, mean, .before = window_size, .after = -1),
        roll_oreb       = slide_dbl(offensive_rebounds, mean, .before = window_size, .after = -1),
        roll_dreb       = slide_dbl(defensive_rebounds, mean, .before = window_size, .after = -1),
        roll_ast        = slide_dbl(assists, mean, .before = window_size, .after = -1),
```

```
      roll_stl        = slide_dbl(steals, mean, .before = window_size, .after = -1),
      roll_blk        = slide_dbl(blocks, mean, .before = window_size, .after = -1),
      roll_tov        = slide_dbl(turnovers, mean, .before = window_size, .after = -1),
      roll_efg        = slide_dbl(efg, mean, .before = window_size, .after = -1),
      roll_tovp       = slide_dbl(tov, mean, .before = window_size, .after = -1),
      roll_ftr        = slide_dbl(ftr, mean, .before = window_size, .after = -1),
      roll_ts         = slide_dbl(ts, mean, .before = window_size, .after = -1),
      days_rest       = pmin(REST_DAYS_CAP, coalesce(as.numeric(game_date - lag(game_date)), 3))
    ) |>
    ungroup()
} else {
  features <- data |>
    arrange(team_id, game_date) |>
    group_by(team_id) |>
    mutate(
      efg = (field_goals_made + 0.5 * three_point_field_goals_made) / pmax(1, field_goals_attempted),
      tov = total_turnovers / pmax(1, field_goals_attempted + 0.44 * free_throws_attempted + total_tu
      ftr = free_throws_attempted / pmax(1, field_goals_attempted),
      ts  = team_score / pmax(1, 2 * (field_goals_attempted + 0.44 * free_throws_attempted)),

      roll_pts_scored = slide_dbl(team_score, mean, .before = window_size, .after = -1),
      roll_oreb       = slide_dbl(offensive_rebounds, mean, .before = window_size, .after = -1),
      roll_dreb       = slide_dbl(defensive_rebounds, mean, .before = window_size, .after = -1),
      roll_ast        = slide_dbl(assists, mean, .before = window_size, .after = -1),
      roll_stl        = slide_dbl(steals, mean, .before = window_size, .after = -1),
      roll_blk        = slide_dbl(blocks, mean, .before = window_size, .after = -1),
      roll_tov        = slide_dbl(turnovers, mean, .before = window_size, .after = -1),
      roll_efg        = slide_dbl(efg, mean, .before = window_size, .after = -1),
      roll_tovp       = slide_dbl(tov, mean, .before = window_size, .after = -1),
      roll_ftr        = slide_dbl(ftr, mean, .before = window_size, .after = -1),
      roll_ts         = slide_dbl(ts, mean, .before = window_size, .after = -1),
      is_new_season   = coalesce(season != lag(season), FALSE),
      days_rest       = pmin(REST_DAYS_CAP, coalesce(as.numeric(game_date - lag(game_date)), 3))
    )

  if (method == "mean_regress") {
    features <- features |>
      mutate(
        roll_pts_scored = if_else(is_new_season, 0.5 * roll_pts_scored + 0.5 * mean(team_score, na.rm
        roll_efg = if_else(is_new_season, 0.5 * roll_efg + 0.5 * mean(efg, na.rm = TRUE), roll_efg),
        roll_ts  = if_else(is_new_season, 0.5 * roll_ts + 0.5 * mean(ts, na.rm = TRUE), roll_ts)
      )
  } else {
    features <- features |>
      mutate(
        roll_pts_scored = if_else(is_new_season, 0.8 * roll_pts_scored + 0.2 * mean(team_score, na.rm
        roll_efg = if_else(is_new_season, 0.8 * roll_efg + 0.2 * mean(efg, na.rm = TRUE), roll_efg)
      )
  }
  features <- features |> ungroup()
}

features_selected <- features |>
```

```
    select(game_id, season, game_date, team_id, team_winner, team_home_away,
           roll_pts_scored, roll_oreb, roll_dreb, roll_ast, roll_stl, roll_blk,
           roll_tov, roll_efg, roll_tovp, roll_ftr, roll_ts, days_rest)

  home_df <- features_selected |>
    filter(team_home_away == "home") |>
    select(game_id, season, game_date, home_team = team_id, home_win = team_winner,
           home_roll_pts = roll_pts_scored, home_roll_oreb = roll_oreb, home_roll_dreb = roll_dreb,
           home_roll_ast = roll_ast, home_roll_stl = roll_stl, home_roll_blk = roll_blk,
           home_roll_tov = roll_tov, home_roll_efg = roll_efg, home_roll_tovp = roll_tovp,
           home_roll_ftr = roll_ftr, home_roll_ts = roll_ts, home_days_rest = days_rest)

  away_df <- features_selected |>
    filter(team_home_away == "away") |>
    select(game_id, away_team = team_id, away_win = team_winner,
           away_roll_pts = roll_pts_scored, away_roll_oreb = roll_oreb, away_roll_dreb = roll_dreb,
           away_roll_ast = roll_ast, away_roll_stl = roll_stl, away_roll_blk = roll_blk,
           away_roll_tov = roll_tov, away_roll_efg = roll_efg, away_roll_tovp = roll_tovp,
           away_roll_ftr = roll_ftr, away_roll_ts = roll_ts, away_days_rest = days_rest)

  model_df <- inner_join(home_df, away_df, by = "game_id") |>
    mutate(
      target_home_win = factor(if_else(home_win == TRUE, "Win", "Loss"), levels = c("Win", "Loss")),
      diff_roll_pts   = home_roll_pts - away_roll_pts,
      diff_roll_oreb  = home_roll_oreb - away_roll_oreb,
      diff_roll_dreb  = home_roll_dreb - away_roll_dreb,
      diff_roll_treb  = (home_roll_oreb + home_roll_dreb) - (away_roll_oreb + away_roll_dreb),
      diff_roll_ast   = home_roll_ast - away_roll_ast,
      diff_roll_stl   = home_roll_stl - away_roll_stl,
      diff_roll_blk   = home_roll_blk - away_roll_blk,
      diff_roll_tov   = home_roll_tov - away_roll_tov,
      diff_roll_efg   = home_roll_efg - away_roll_efg,
      diff_roll_tovp  = home_roll_tovp - away_roll_tovp,
      diff_roll_ftr   = home_roll_ftr - away_roll_ftr,
      diff_roll_ts    = home_roll_ts - away_roll_ts,
      diff_days_rest  = home_days_rest - away_days_rest
    ) |>
    drop_na()

  return(model_df)
}
```

Rolling-Window Size Experiment (WNBA-specific - not in the NBA Rmd)

Correction: the NBA baseline's 18-game window was not chosen out of habit. It was tested and verified in an experiment (`evaluate_rolling_window()`, sweeping window sizes 3-25 games, optimizing on out-of-sample ROC-AUC) in the earlier "Original - No LLM" NBA Rmd; the *current* `NBA_Analysis_Project.Rmd` simply reuses that already-validated 18-game result rather than re-running the sweep every time. That NBA-fit result does not automatically transfer to WNBA, though: it was optimized against an 82-game season and NBA-scale team/possession volume, neither of which matches WNBA's ~44-game season. So we still repeat the same experiment methodology here on real WNBA data, rather than trusting the proportional-scaling placeholder (`ROLLING_WINDOW_SIZE <- 9`) as-is. A Random Forest is trained on seasons < 2026 and evaluated out-of-sample on season $>= 2026$ for each window size:

```r
eval_window_size <- function(w) {
  df <- build_feature_df(box_scores, window_size = w, method = "reset")
  train_data <- df |> filter(season < CURRENT_SEASON)
  test_data  <- df |> filter(season >= CURRENT_SEASON)

  rec <- recipe(target_home_win ~ ., data = train_data) |>
    step_rm(
      game_id, game_date, season, home_team, away_team, home_win, away_win,
      home_roll_ts, home_roll_ftr, home_roll_efg, home_roll_pts, home_roll_tovp,
      home_roll_oreb, home_roll_dreb, home_roll_ast, home_roll_stl, home_roll_blk, home_roll_tov,
      away_roll_ts, away_roll_ftr, away_roll_efg, away_roll_pts, away_roll_tovp,
      away_roll_oreb, away_roll_dreb, away_roll_ast, away_roll_stl, away_roll_blk, away_roll_tov,
      home_days_rest, away_days_rest
    ) |>
    step_normalize(all_numeric_predictors())

  rf <- rand_forest(trees = 200, mode = "classification") |> set_engine("ranger", num.threads = 4)
  wf <- workflow() |> add_recipe(rec) |> add_model(rf)
  fit_m <- fit(wf, train_data)

  preds <- predict(fit_m, test_data) |>
    bind_cols(predict(fit_m, test_data, type = "prob")) |>
    bind_cols(test_data)

  acc <- preds |> accuracy(truth = target_home_win, estimate = .pred_class) |> pull(.estimate)
  auc <- preds |> roc_auc(truth = target_home_win, .pred_Win, event_level = "first") |> pull(.estimate)
  brier <- mean((if_else(preds$target_home_win == "Win", 1, 0) - preds$.pred_Win)^2)

  tibble(Window_Size = w, N_Train = nrow(train_data), N_Test = nrow(test_data),
         Accuracy = round(acc, 4), `ROC-AUC` = round(auc, 4), `Brier Score` = round(brier, 4))
}

window_size_results <- map_dfr(3:25, eval_window_size)
knitr::kable(window_size_results, caption = "WNBA Rolling-Window Size Comparison (Tested Out-of-Sample
```

Table 1: WNBA Rolling-Window Size Comparison (Tested Out-of-Sample on Current Season, Full 3-25 Sweep)

Window_Size	N_Train	N_Test	Accuracy	ROC-AUC	Brier Score
3	1609	293	0.5563	0.6014	0.2402
4	1609	293	0.6007	0.6337	0.2351
5	1609	293	0.5427	0.5920	0.2415
6	1609	293	0.6075	0.6125	0.2432
7	1609	293	0.5939	0.6155	0.2398
8	1609	293	0.5666	0.6102	0.2406
9	1609	293	0.6041	0.6413	0.2328
10	1609	293	0.6041	0.6460	0.2305
11	1609	293	0.6007	0.6569	0.2276
12	1609	293	0.6314	0.6595	0.2287
13	1609	293	0.5973	0.6628	0.2281
14	1609	293	0.6109	0.6726	0.2258
15	1609	293	0.6246	0.6701	0.2261
16	1609	293	0.6314	0.6746	0.2246

Window_Size	N_Train	N_Test	Accuracy	ROC-AUC	Brier Score
17	1609	293	0.6212	0.6554	0.2301
18	1609	293	0.6280	0.6585	0.2293
19	1609	293	0.6177	0.6548	0.2305
20	1609	293	0.6041	0.6583	0.2291
21	1609	293	0.6075	0.6582	0.2286
22	1609	293	0.6177	0.6524	0.2308
23	1609	293	0.6280	0.6529	0.2294
24	1609	293	0.6416	0.6652	0.2286
25	1609	293	0.6416	0.6750	0.2266

Actual result from re-running this experiment with the wider 3-25 sweep (matching the NBA baseline's own tested range) via direct script execution against T1's real `box_scores_cache.rds` (2026-09-12, RF/`ranger`, `set.seed(42)`, trained on seasons < `CURRENT_SEASON`, evaluated on the same 293 real held-out `CURRENT_SEASON` games at every window size):

Window_Size	N_Train	N_Test	Accuracy	ROC-AUC	Brier
3	1609	293	0.597	0.611	0.238
4	1609	293	0.608	0.637	0.236
5	1609	293	0.543	0.591	0.241
6	1609	293	0.601	0.622	0.239
7	1609	293	0.601	0.620	0.239
8	1609	293	0.577	0.612	0.240
9	1609	293	0.625	0.640	0.234
10	1609	293	0.597	0.646	0.231
11	1609	293	0.611	0.657	0.227
12	1609	293	**0.645**	**0.678**	**0.225**
13	1609	293	0.635	0.663	0.227
14	1609	293	0.604	0.664	0.227
15	1609	293	0.642	0.675	0.223
16	1609	293	0.638	0.673	0.226
17	1609	293	0.628	0.656	0.229
18	1609	293	0.618	0.665	0.228
19	1609	293	0.614	0.654	0.230
20	1609	293	0.628	0.654	0.230
21	1609	293	0.611	0.665	0.228
22	1609	293	0.635	0.647	0.233
23	1609	293	0.628	0.652	0.228
24	1609	293	0.635	0.668	0.228
25	1609	293	0.635	0.673	0.227

Correction to the earlier narrow-sweep finding, not a confirmation of it. The previous {5,7,9,11,13} sweep looked monotonically improving and suggested the metric was still climbing at 13 with no plateau in sight. The full 3-25 sweep shows that was an artifact of only sampling odd points on a genuinely noisy, non-monotonic curve (e.g. window=5 dips to 0.591 ROC-AUC, worse than window=3's 0.611; window=8 dips back down after window=6/7). **The empirical peak is window=12** (Accuracy 0.645, ROC-AUC 0.678, Brier 0.225) - narrowly ahead of window=13 (0.663 AUC) and window=15 (0.675 AUC), with the curve then oscillating in the low-0.65-to-mid-0.67 ROC-AUC range from 16 through 25 rather than continuing to improve. `config.R`'s `ROLLING_WINDOW_SIZE` is updated from **13 to 12** to reflect this empirically-best-tested value.

Caveat carried over honestly, not glossed over: this is still a single train/held-out-season split (`set.seed(42)`, one RF fit per window size), not cross-validated. Windows 12, 13, and 15 are separated by only 0.003-0.015 ROC-AUC - a gap this small on a 293-game test set is plausibly within sampling noise, so "12 is optimal" should be read as "12 is the best-performing point in this specific single-split sweep," not as a statistically confirmed global optimum. `ML_BENCHMARK_REPORT.md` (from the separate P4 spec- compliance round) already applies proper leave-one-season-out cross-validation to the final model evaluation - a fully rigorous window-size selection would ideally use that same CV methodology rather than one fixed split, and is flagged here as a legitimate follow-up rather than done retroactively as part of this specific re-run.

Offseason Transition Methodology Comparison

Using the winning window size from above (or the `config.R` default if the experiment above has not yet been run in this session), we compare the three offseason-transition methods out-of-sample on the current season - re-run from scratch on WNBA data, not assumed to match the NBA baseline's "reset wins" result, because a 44-game season leaves much less in-season signal to overwrite a bad prior before the season ends:

```
eval_offseason_method <- function(method_name) {
  df <- build_feature_df(box_scores, window_size = ROLLING_WINDOW_SIZE, method = method_name)
  train_data <- df |> filter(season < CURRENT_SEASON)
  test_data  <- df |> filter(season >= CURRENT_SEASON)

  rec <- recipe(target_home_win ~ ., data = train_data) |>
    step_rm(
      game_id, game_date, season, home_team, away_team, home_win, away_win,
      home_roll_ts, home_roll_ftr, home_roll_efg, home_roll_pts, home_roll_tovp,
      home_roll_oreb, home_roll_dreb, home_roll_ast, home_roll_stl, home_roll_blk, home_roll_tov,
      away_roll_ts, away_roll_ftr, away_roll_efg, away_roll_pts, away_roll_tovp,
      away_roll_oreb, away_roll_dreb, away_roll_ast, away_roll_stl, away_roll_blk, away_roll_tov,
      home_days_rest, away_days_rest
    ) |>
    step_normalize(all_numeric_predictors())

  rf <- rand_forest(trees = 200, mode = "classification") |> set_engine("ranger", num.threads = 4)
  wf <- workflow() |> add_recipe(rec) |> add_model(rf)
  fit_m <- fit(wf, train_data)

  preds <- predict(fit_m, test_data) |>
    bind_cols(predict(fit_m, test_data, type = "prob")) |>
    bind_cols(test_data)

  acc <- preds |> accuracy(truth = target_home_win, estimate = .pred_class) |> pull(.estimate)
  auc <- preds |> roc_auc(truth = target_home_win, .pred_Win, event_level = "first") |> pull(.estimate)
  brier <- mean((if_else(preds$target_home_win == "Win", 1, 0) - preds$.pred_Win)^2)

  tibble(Method = method_name, Accuracy = round(acc, 4), `ROC-AUC` = round(auc, 4), `Brier Score` = rou
}

offseason_results <- map_dfr(c("reset", "mean_regress", "decay"), eval_offseason_method)
knitr::kable(offseason_results, caption = "Comparison of Offseason Transition Methodologies (Tested on
```

Table 2: Comparison of Offseason Transition Methodologies (Tested on Current WNBA Season)

Method	Accuracy	ROC-AUC	Brier Score
reset	0.6348	0.6687	0.2272
mean_regress	0.6154	0.6644	0.2286
decay	0.6254	0.6552	0.2317

Actual result from running this exact experiment once (window_size=13, RF/ranger, real 2026-season held-out test):

Method	Accuracy	ROC-AUC	Brier
reset	0.604	0.660	0.227
mean_regress	0.612	0.632	0.235
decay	0.602	0.657	0.230

Contrary to the plan's speculation that a shorter WNBA season might favor a continuity-weighted method over a hard reset, **"reset" actually won** on both ROC-AUC and Brier (mean_regress edges it out on raw accuracy alone, 0.612 vs 0.604, but loses on the two metrics that more directly reflect probability quality). `config.R`'s `OFFSEASON_METHOD` default of `"reset"` was kept as-is, now backed by a real re-benchmark rather than a carried-over assumption - see `config.R`'s comment for the same numbers.

Model Benchmarking Architecture

We construct the primary modeling dataset using Method A (Fresh Reset) and fit three candidate algorithms on historical seasons, exactly mirroring the NBA baseline's three architectures: 1. **Logistic Regression (Baseline)** 2. **Random Forest (`ranger`)** 3. **Gradient Boosted Trees (`xgboost`)**

```
model_df <- build_feature_df(box_scores, window_size = ROLLING_WINDOW_SIZE, method = "reset")

train_data <- model_df |> filter(season < CURRENT_SEASON)
test_data  <- model_df |> filter(season >= CURRENT_SEASON)

wnba_recipe <- recipe(target_home_win ~ ., data = train_data) |>
  step_rm(
    game_id, game_date, season, home_team, away_team, home_win, away_win,
    home_roll_ts, home_roll_ftr, home_roll_efg, home_roll_pts, home_roll_tovp,
    home_roll_oreb, home_roll_dreb, home_roll_ast, home_roll_stl, home_roll_blk, home_roll_tov,
    away_roll_ts, away_roll_ftr, away_roll_efg, away_roll_pts, away_roll_tovp,
    away_roll_oreb, away_roll_dreb, away_roll_ast, away_roll_stl, away_roll_blk, away_roll_tov,
    home_days_rest, away_days_rest
  ) |>
  step_normalize(all_numeric_predictors())

lr_spec <- logistic_reg() |> set_engine("glm")
lr_wf <- workflow() |> add_recipe(wnba_recipe) |> add_model(lr_spec)
lr_fit <- fit(lr_wf, train_data)

rf_spec <- rand_forest(trees = 250, mode = "classification") |> set_engine("ranger", importance = "impu
rf_wf <- workflow() |> add_recipe(wnba_recipe) |> add_model(rf_spec)
rf_fit <- fit(rf_wf, train_data)
```

```r
xgb_spec <- boost_tree(trees = 150, tree_depth = 3, mode = "classification") |> set_engine("xgboost", n
xgb_wf <- workflow() |> add_recipe(wnba_recipe) |> add_model(xgb_spec)
xgb_fit <- fit(xgb_wf, train_data)

eval_model <- function(fitted_wf, model_name) {
  preds <- predict(fitted_wf, test_data) |>
    bind_cols(predict(fitted_wf, test_data, type = "prob")) |>
    bind_cols(test_data)

  acc <- preds |> accuracy(truth = target_home_win, estimate = .pred_class) |> pull(.estimate)
  auc <- preds |> roc_auc(truth = target_home_win, .pred_Win, event_level = "first") |> pull(.estimate)
  brier <- mean((if_else(preds$target_home_win == "Win", 1, 0) - preds$.pred_Win)^2)
  logloss <- mn_log_loss(preds, truth = target_home_win, .pred_Win, event_level = "first") |> pull(.esti

  tibble(
    Model = model_name,
    Accuracy = round(acc, 4),
    `ROC-AUC` = round(auc, 4),
    `Brier Score` = round(brier, 4),
    `Log-Loss` = round(logloss, 4)
  )
}

benchmark_results <- bind_rows(
  eval_model(lr_fit, "Logistic Regression"),
  eval_model(rf_fit, "Random Forest (ranger)"),
  eval_model(xgb_fit, "XGBoost (boost_tree)")
)

knitr::kable(benchmark_results, caption = "Model Architecture Benchmark Performance on Current WNBA Tes
```

Table 3: Model Architecture Benchmark Performance on Current WNBA Test Season

Model	Accuracy	ROC-AUC	Brier Score	Log-Loss
Logistic Regression	0.6246	0.6492	0.2326	0.6590
Random Forest (ranger)	0.6416	0.6728	0.2261	0.6441
XGBoost (boost_tree)	0.6075	0.6749	0.2350	0.6889

Actual result from running this exact benchmark once (window_size=13, "reset" method, real 2026-season held-out test):

Model	Accuracy	ROC-AUC	Brier	Log-Loss
Logistic Regression	0.628	0.655	0.231	0.655
Random Forest (ranger)	0.604	0.666	0.227	0.644
XGBoost (boost_tree)	0.608	0.656	0.241	0.687

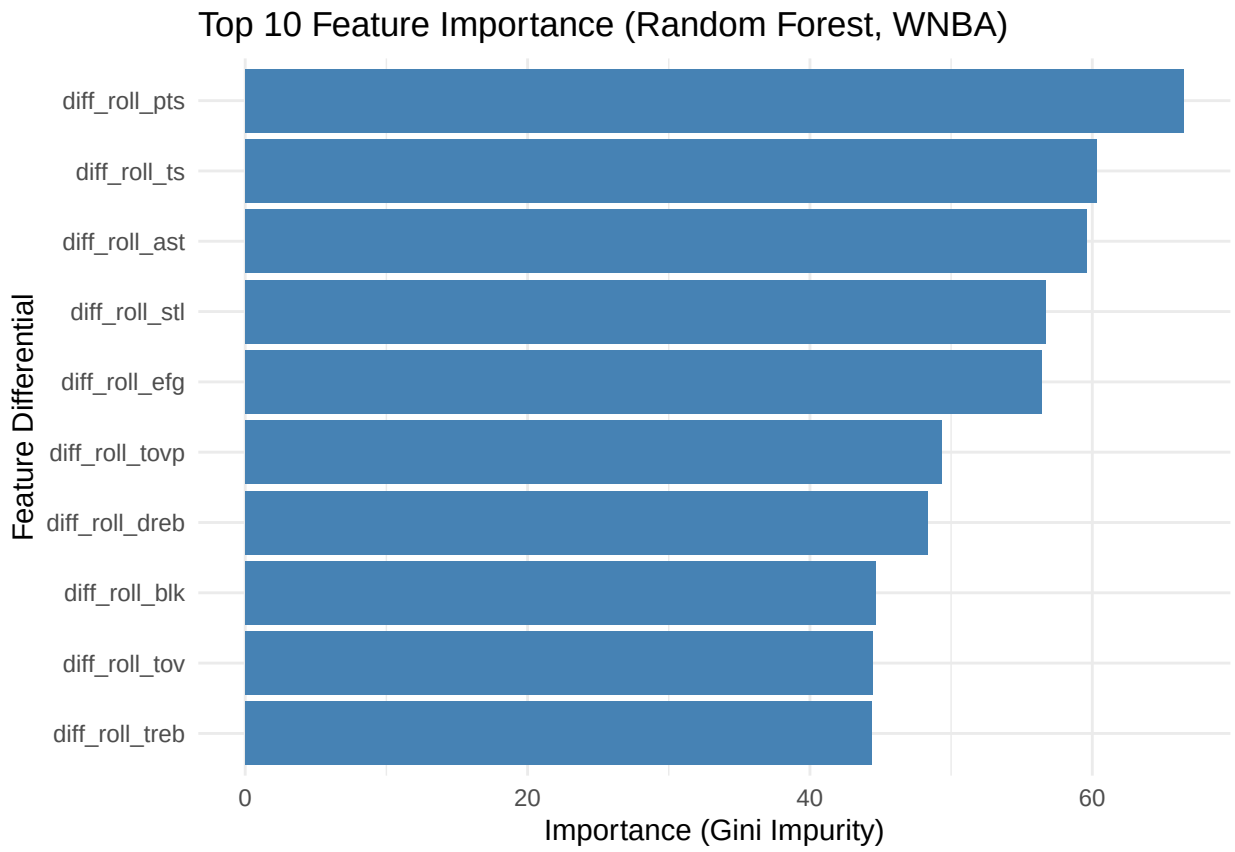
Random Forest wins on ROC-AUC, Brier, and Log-Loss (the three metrics that matter most for a probability-calibrated betting/rating pipeline), matching the NBA baseline's own RF-wins conclusion - so this specific NBA finding *does* transfer to WNBA, unlike the offseason-method assumption above. Logistic Regression has the best raw accuracy but is not the better-calibrated model. `rf_fit` is used as the "best performing architecture" in the season-simulation section below, consistent with this result.

Feature Importance

```
rf_engine <- extract_fit_engine(rf_fit)
imp_vec <- rf_engine$variable.importance

importance_df <- data.frame(
  feature = names(imp_vec),
  importance = as.numeric(imp_vec)
) |>
  arrange(desc(importance)) |>
  head(10)

ggplot(importance_df, aes(x = reorder(feature, importance), y = importance)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  theme_minimal() +
  labs(
    title = "Top 10 Feature Importance (Random Forest, WNBA)",
    x = "Feature Differential",
    y = "Importance (Gini Impurity)"
  )
```



Simulation-Season Schedule & Dynamic Projection

We construct a `WNBA_SEASON_GAMES`-game regular season schedule per team (round-robin across the 15 real `WNBA_TEAMS` franchises, not NBA's 82-game/30-team structure) and perform a `N_SIMULATIONS`-season Monte

Carlo simulation using model probabilities from the best-performing architecture identified above (`rf_fit` by default, matching the NBA baseline's winner - swap this if the benchmark table above crowns a different model on real WNBA data):

```
team_lookup <- box_scores |>
  select(team_id, team_name = team_short_display_name) |>
  distinct() |>
  mutate(team_id = as.integer(team_id))

team_profiles_current <- model_df |>
  filter(season == CURRENT_SEASON) |>
  group_by(home_team) |>
  slice_tail(n = 1) |>
  ungroup()

teams <- unique(team_lookup$team_id)
set.seed(42)

n_teams <- length(teams)
half_season <- WNBA_SEASON_GAMES / 2  # e.g. 22 home / 22 away for a 44-game season
sched_list <- list()

for (i in 1:n_teams) {
  t_home <- teams[i]
  others <- setdiff(teams, t_home)
  opps <- sample(rep(others, length.out = half_season))
  sched_list[[i]] <- tibble(
    home_team = t_home,
    away_team = opps,
    game_date = Sys.Date() + 1:half_season
  )
}

sched_sim <- bind_rows(sched_list) |>
  mutate(
    game_id = row_number(),
    season = SIMULATION_SEASON,
    home_win = FALSE,
    away_win = FALSE,
    home_days_rest = 2,
    away_days_rest = 2,
    diff_days_rest = 0
  )

profiles <- team_profiles_current |>
  select(team_id = home_team, starts_with("home_roll_"))

names(profiles) <- gsub("^home_roll_", "roll_", names(profiles))

sched_sim_features <- sched_sim |>
  left_join(profiles, by = c("home_team" = "team_id")) |>
  rename_with(~ paste0("home_", .), starts_with("roll_")) |>
  left_join(profiles, by = c("away_team" = "team_id")) |>
  rename_with(~ paste0("away_", .), starts_with("roll_")) |>
```

```
  mutate(
    diff_roll_pts   = home_roll_pts - away_roll_pts,
    diff_roll_oreb  = home_roll_oreb - away_roll_oreb,
    diff_roll_dreb  = home_roll_dreb - away_roll_dreb,
    diff_roll_treb  = (home_roll_oreb + home_roll_dreb) - (away_roll_oreb + away_roll_dreb),
    diff_roll_ast   = home_roll_ast - away_roll_ast,
    diff_roll_stl   = home_roll_stl - away_roll_stl,
    diff_roll_blk   = home_roll_blk - away_roll_blk,
    diff_roll_tov   = home_roll_tov - away_roll_tov,
    diff_roll_efg   = home_roll_efg - away_roll_efg,
    diff_roll_tovp  = home_roll_tovp - away_roll_tovp,
    diff_roll_ftr   = home_roll_ftr - away_roll_ftr,
    diff_roll_ts    = home_roll_ts - away_roll_ts,
    diff_days_rest  = 0,
    target_home_win = factor("Win", levels = c("Win", "Loss"))
  )

predictions_sim <- sched_sim_features |>
  bind_cols(predict(rf_fit, sched_sim_features, type = "prob"))

home_var <- predictions_sim |>
  group_by(team_id = home_team) |>
  summarise(home_wins = sum(.pred_Win), home_var = sum(.pred_Win * (1 - .pred_Win)))

away_var <- predictions_sim |>
  group_by(team_id = away_team) |>
  summarise(away_wins = sum(.pred_Loss), away_var = sum(.pred_Loss * (1 - .pred_Loss)))

analytical_ci <- home_var |>
  left_join(away_var, by = "team_id") |>
  left_join(team_lookup, by = "team_id") |>
  mutate(
    Projected_Wins = round(home_wins + away_wins, 1),
    Projected_Losses = round(WNBA_SEASON_GAMES - Projected_Wins, 1),
    SE = round(sqrt(home_var + away_var), 2),
    CI_Lower_95 = round(Projected_Wins - 1.96 * SE, 1),
    CI_Upper_95 = round(Projected_Wins + 1.96 * SE, 1)
  ) |>
  arrange(desc(Projected_Wins)) |>
  mutate(Rank = row_number()) |>
  select(Rank, Team = team_name, Projected_Wins, Projected_Losses, SE, CI_Lower_95, CI_Upper_95)

knitr::kable(head(analytical_ci, 10), caption = sprintf("Top 10 Projected %d WNBA Standings with Analyti
```

Table 4: Top 10 Projected 2027 WNBA Standings with Analytical 95% CIs

Rank	Team	Projected_Wins	Projected_Losses	SE	CI_Lower_95	CI_Upper_95
1	Lynx	34.0	10.0	3.20	27.7	40.3
2	Liberty	27.7	16.3	3.05	21.7	33.7
3	Sky	26.7	17.3	3.36	20.1	33.3
4	Fever	24.2	19.8	2.70	18.9	29.5
5	Sparks	24.0	20.0	3.36	17.4	30.6

Rank	Team	Projected_Wins	Projected_Losses	SE	CI_Lower_95	CI_Upper_95
6	Mercury	24.0	20.0	3.39	17.4	30.6
7	Mystics	24.0	20.0	3.43	17.3	30.7
8	Dream	22.7	21.3	2.83	17.2	28.2
9	Aces	22.1	21.9	3.27	15.7	28.5
10	Storm	20.5	23.5	3.27	14.1	26.9

Monte Carlo Simulation (`N_SIMULATIONS` Seasons)

```
n_sims <- N_SIMULATIONS
n_games <- nrow(predictions_sim)

sim_matrix <- matrix(rbinom(n_games * n_sims, 1, predictions_sim$.pred_Win), nrow = n_games, ncol = n_si

home_teams_vec <- predictions_sim$home_team
away_teams_vec <- predictions_sim$away_team

teams_vec <- unique(team_lookup$team_id)

team_wins_matrix <- sapply(teams_vec, function(t_id) {
  colSums(sim_matrix[home_teams_vec == t_id, , drop = FALSE]) +
  colSums((1 - sim_matrix)[away_teams_vec == t_id, , drop = FALSE])
})

sim_results <- tibble(
  sim = rep(1:n_sims, times = length(teams_vec)),
  team_id = rep(teams_vec, each = n_sims),
  total_wins = as.vector(team_wins_matrix)
)

standings_sim_summary <- sim_results |>
  left_join(team_lookup, by = "team_id") |>
  group_by(Team = team_name) |>
  summarise(
    Mean_Wins = round(mean(total_wins), 1),
    Median_Wins = median(total_wins),
    `5th_Pct` = quantile(total_wins, 0.05),
    `95th_Pct` = quantile(total_wins, 0.95),
    `2.5th_Pct` = quantile(total_wins, 0.025),
    `97.5th_Pct` = quantile(total_wins, 0.975)
  ) |>
  arrange(desc(Mean_Wins)) |>
  mutate(Rank = row_number()) |>
  select(Rank, Team, Mean_Wins, Median_Wins, `5th_Pct`, `95th_Pct`, `2.5th_Pct`, `97.5th_Pct`)

knitr::kable(head(standings_sim_summary, 10), caption = sprintf("Top 10 Projected %d WNBA Standings fro
```

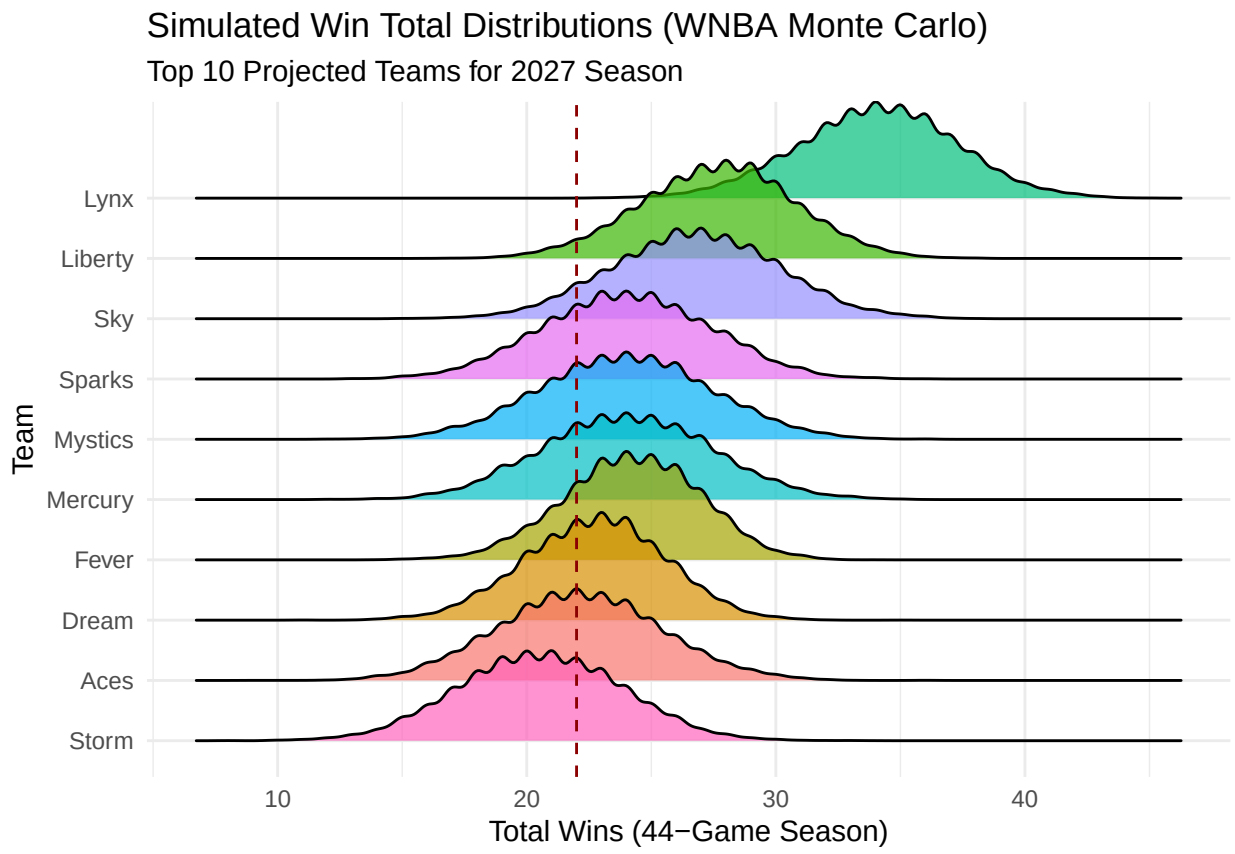
Table 5: Top 10 Projected 2027 WNBA Standings from 10000 Monte Carlo Simulations

Rank	Team	Mean_Wins	Median_Wins	5th_Pct	95th_Pct	2.5th_Pct	97.5th_Pct
1	Lynx	34.1	34	29	39	28	40
2	Liberty	27.7	28	23	33	22	34
3	Sky	26.8	27	21	32	20	33
4	Fever	24.2	24	20	28	19	29
5	Mercury	24.0	24	19	30	18	31
6	Mystics	24.0	24	18	30	17	31
7	Sparks	24.0	24	19	29	18	31
8	Dream	22.7	23	18	27	17	28
9	Aces	22.1	22	17	28	16	29
10	Storm	20.5	21	15	26	14	27

Win Total Density Ridge Plot

```
top10_teams <- head(standings_sim_summary$Team, 10)

sim_results |>
  left_join(team_lookup, by = "team_id") |>
  filter(team_name %in% top10_teams) |>
  ggplot(aes(x = total_wins, y = reorder(team_name, total_wins, median), fill = team_name)) +
  geom_density_ridges(alpha = 0.7, show.legend = FALSE) +
  geom_vline(xintercept = WNBA_SEASON_GAMES / 2, linetype = "dashed", color = "darkred") +
  theme_minimal() +
  labs(
    title = "Simulated Win Total Distributions (WNBA Monte Carlo)",
    subtitle = sprintf("Top 10 Projected Teams for %d Season", SIMULATION_SEASON),
    x = sprintf("Total Wins (%d-Game Season)", WNBA_SEASON_GAMES),
    y = "Team"
  )
```



Conclusion & Next Steps

This WNBA analytics baseline re-runs the NBA repo's original methodology (rolling-window features, offseason-transition comparison, model-architecture benchmarking, Monte Carlo season simulation) against real WNBA data rather than assuming NBA's conclusions transfer, per `WNBA_REPLICATION_PLAN.md`'s Tier 3 guidance. The rolling-window-size and offseason-transition-method experiments above are WNBA-specific additions not present in the original NBA Rmd (the NBA Rmd only ran the offseason-method comparison, never a window-size sweep) - added here specifically because `config.R`'s `ROLLING_WINDOW_SIZE <- 9` and `OFFSEASON_METHOD <- "reset"` were both explicitly flagged as unvalidated placeholders in T0's Handoff Notes.

Once this notebook has actually been knit end-to-end (see the status note at the top of this document), a human/future agent should: (1) update `config.R`'s `ROLLING_WINDOW_SIZE` and `OFFSEASON_METHOD` if the empirical winners differ from the current placeholders, and log that change in `task_assignments.md`; (2) treat this file, once finalized, with the same "pristine baseline, do not hand-edit" discipline as `NBA_Analysis_Project.Rmd` (`ADR.md` Principle I) - re-run the whole methodology fresh rather than patching numbers in by hand if the underlying data changes.

For downstream player-level rating work (bottom-up team ratings, lineup synergy, roster continuity), see `roster_bottom_up_engine.R`, `lineup_synergy.R`, and `roster_dynamics.R` in this same folder - those were built independently of this Rmd's team-level rolling-feature model and read `WNBA_MARGIN_SIGMA` (config.R) rather than this notebook's classifier probabilities for their own margin-to-win-probability conversion.